ZCAS UNIVERSITY

SCHOOL OF COMPUTING AND APPLIED SCIENCES

CIT2371 DATA SECURITY

End of Semester Final Examination

Monday 8th June 2026

14:00 – 17:00 HOURS

TIME ALLOWED: **WRITING – THREE (3) HOURS**

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Section B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer the following question.

QUESTION 1.

Indeco Corporation is a leading multinational company that deals with sensitive customer information, including personal and financial data. The company has a vast database that stores this information and relies on it to carry out its operations effectively. However, recently, Indeco Corporation experienced a significant security breach, resulting in the compromise of thousands of customer records. This incident has raised concerns among customers and stakeholders, and the company is now faced with the task of addressing these security issues to restore trust and ensure the safety of their information storage.

Required:

a) Provide measures Indeco Corporation can implement to enhance the physical security of their information storage facilities. **(8 Marks)**

b) Explain how Indeco Corporation ensures the security of data transmission between their various systems and locations? **(8 Marks)**

c) Discuss the strategies Indeco Corporation employs to protect their stored data from cyber threats such as hacking and malware attacks. **(8 Marks)**

d) Explain ways Indeco Corporation ensures the integrity and authenticity of their stored information. **(8 Marks)**

e) Give the steps Indeco Corporation would take to ensure data backup and disaster recovery. **(8 Marks)**

[Total 40 Marks]

SECTION B

Choose any three questions from this section.

QUESTION 2.

Jonathan Zozghi is a cryptanalyst working for a top intelligence agency. He has been assigned a challenging task: decrypting a highly encrypted message intercepted from a suspicious group involved in illegal activities. The agency believes that the message contains critical information about their next target.

Jonathan starts by analyzing the encryption algorithm used in the message. After careful examination, he identifies that it is a symmetric key encryption algorithm. He also notices that the message is encrypted using a substitution cipher. However, this substitution cipher is not a straightforward one, as the group has applied various complex transformations to make it more difficult to crack.

Jonathan realizes that she needs to apply different techniques and methods to break the encryption. He begins by looking for patterns and common words in the message. By analyzing the frequency of letters and letter pairs, she identifies a few probable substitutions. He starts mapping these substitutions and gradually builds a partial decryption key.

Required:

a) Explain how Jonathan would leverage frequency analysis to break the encryption.

(5 Marks)

b) Describe other techniques that Jonathan can employ to decrypt the message.

(5 Marks)

c) Discuss how Jonathan can deal with the complex transformations applied to the substitution cipher during decryption. **(5 Marks)**

d) Discuss the actions that can be taken if Jonathan finds the decryption process too challenging. **(5 Marks)**

[Total 20 Marks]

QUESTION 3.

You happen to work as Cyber security specialist at Liquid Telecom, and you have been invited to take part in an important Annual General Meeting for the Ministry of Home Affairs in Livingstone. You have been asked to prepare yourself to explain important concepts related to cryptography.

Required:

Discuss the following:

a) Feistel Cipher **(5 Marks)**

b) Public Encryption **(5 Marks)**

c) Public Key Infrastructure (PKI): How it works and its benefits **(10 Marks)**

[Total 20 Marks]

QUESTION 4.

Access Bank Ltd is implementing a new access control system to make sure its sensitive data is secure. The system will restrict access to certain areas of the bank's network and physical premises based on an employee's role and level of authorization. The access control system will be managed by the IT department which will use various ways to restrict access to its data.

Required:

a) Explain the measures Access Bank Ltd would put in place to ensure that only authorized individuals are granted access to sensitive data and areas of its network. **(5 Marks)**

b) Demonstrate how the access control system will ensure that unauthorized individuals cannot bypass the system and gain access to Access Bank Ltd restricted areas. **(5 Marks)**

a) Describe the principle of least privilege in access control at Access Bank Ltd and why you think is it important in cybersecurity. **(5 Marks)**

b) Explain the difference between mandatory access control and discretionary access control. **(5 Marks)**

[Total 20 Marks]

QUESTION 5.

Suppose that Kalulushi Mine Company's server has been compromised, resulting in the loss of critical data. The IT department suspects that an insider was responsible for the breach and requests the assistance of a digital forensics expert to investigate the incident.

Required:

a) Explain the process of acquiring a forensic image from a compromised server and describe the importance of using write-blocking techniques. **(5 Marks)**

b) Describe the significance of analyzing system logs in a digital forensics investigation and discuss how they can provide valuable insights into the activities of an insider. **(5 Marks)**

c) Discuss the role of network forensics in tracing the activities of an insider and explain how it can be used to identify unauthorized connections and data transfers. **(5 Marks)**

d) Explain the concept of user activity monitoring in digital forensics and discuss how it can help investigators track the actions of an insider, including login times, file accesses, and attempts to cover their tracks. **(5 Marks)**

[Total 20 Marks]

END OF EXAMINATION